band wrapped around my finger [A] causes my finger to turn white [B]) then the line connecting A and B makes a 90-degree angle relative to the time slices. If A and B take place at different locations in space, whatever traveled from A to B to exert the influence (my pebble traveling from slingshot to window) did so at less than light speed, which means the angle differs from 90 degrees (the angle when no speed is involved) by less than 45 degrees—i.e. the angle with respect to the time slices (the space axes) is greater than 45 degrees. (Remember from endnote 9 of this chapter that light speed sets the limit and such motion traces out 45-degree lines.) Now, as in endnote 9, the different time slicings associated with an observer in motion are angled relative to those of an observer at rest, but the angle is always *less* than 45 degrees (since the relative motion between two material observers is always less than the speed of light). And since the angle associated with causally related events is always *greater* than 45 degrees, the time slices of an observer, who necessarily travels at less than light speed, cannot first encounter the effect and then later encounter the cause. To all observers, cause will precede effect.

12. The notion that causes precede their effects (see the preceding note) would, among other things, be challenged if influences could travel faster than the speed of light.

13. Isaac Newton, *Sir Isaac Newton's Mathematical Principles of Natural Philosophy and His System of the World,* trans. A. Motte and Florian Cajori (Berkeley: University of California Press, 1962), vol. 1, p. 634.

14. Because the gravitational pull of the earth differs from one location to another, a spatially extended, freely falling observer can still detect a residual gravitational influence. Namely, if the observer, while falling, releases two baseballs—one from his outstretched right arm and the other from his left—each will fall along a path toward the earth's center. So, from the observer's perspective, he will be falling straight down toward the earth's center, while the ball released from his right hand will travel downward and slightly toward the left, while the ball released from his left hand will travel downward and slightly toward the right. Through careful measurement, the observer will therefore see that the distance between the two baseballs slowly decreases; they move